



Characterizing Normal Brain Development via EEG Pattern Matching

An automated pipeline for discovering and comparing EEG motifs across developmental ages 5–21

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FINAL PROJECT – PHASE B
Code: 26–1–R–11

1. WHY THIS STUDY?



Develop an externalised pipeline to discover and compare recurring EEG motifs.



Compare patterns across ages 5–21 to characterise normal brain development.



Generate reproducible, interpretable results for future EEG studies.

2. DATASET (HBN-EEG)



HBN-EEG in BIDS format
EEG set: recordings + participants.tsv



Ages 5–21
Balanced age groups

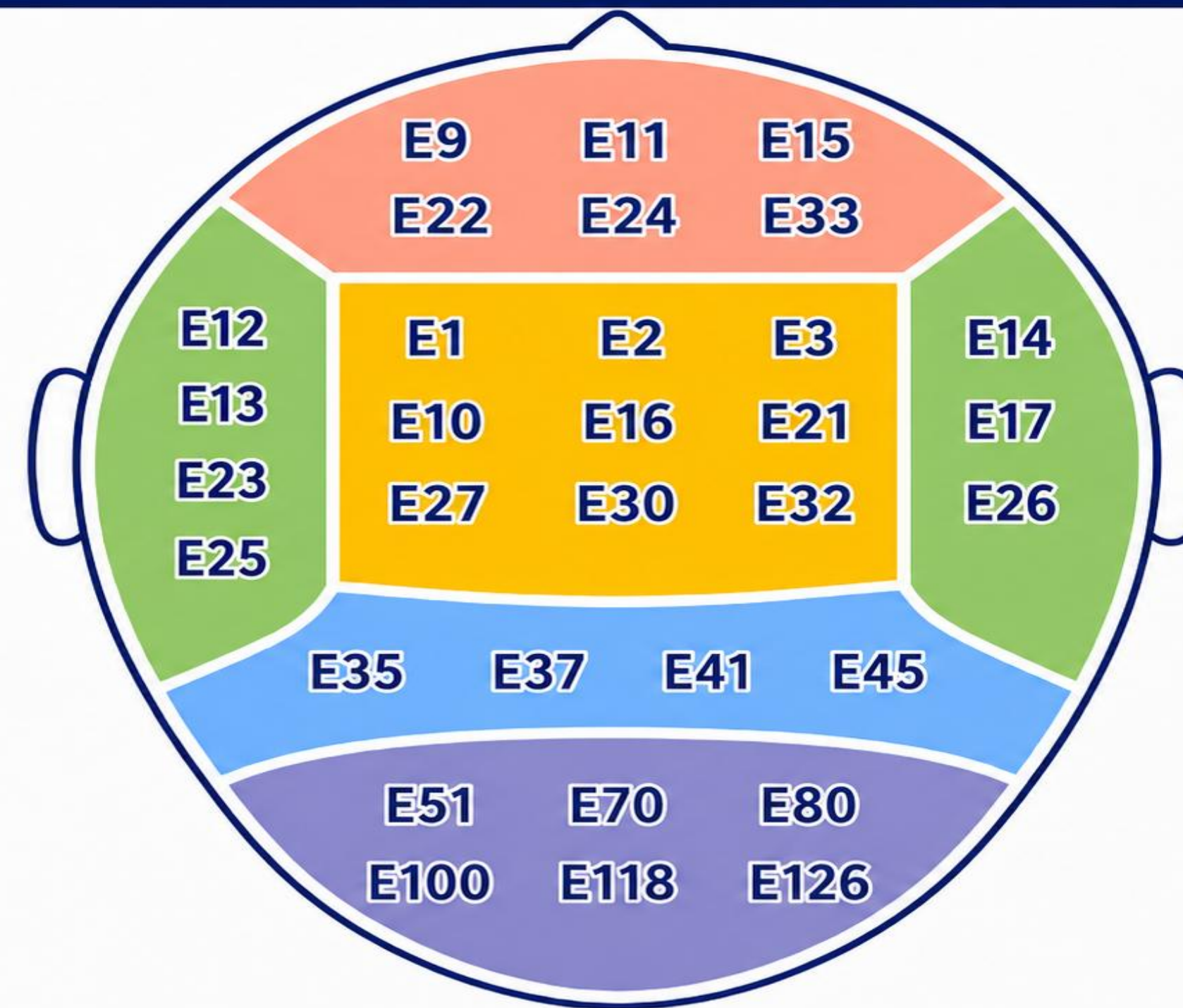


Tasks included
DespicableMe • RestingState • contrastChangeDetection



Participants compared
across balanced age groups

3. REPRESENTATIVE CHANNEL COVERAGE



47 representative electrodes sampled from 5 scalp regions

- Frontal
- Central
- Temporal
- Parietal
- Occipital / Posterior

All 47 Selected Electrodes:
E1, E2, E3, E4, E5, E6, E7, E8, E9, E10, E11, E12, E13, E14, E15, E16, E17, E18, E19, E20, E21, E22, E23, E24, E25, E26, E27, E28, E30, E32, E33, E35, E37, E41, E45, E51, E60, E70, E80, E90, E100, E110, E118, E120, E122, E124, E126

Examples shown in the map.

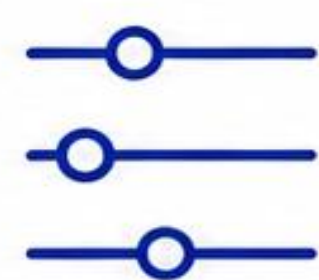
4. AUTOMATED ANALYSIS PIPELINE

1



Load Data
EEG set
from HBN-EEG

2



Preprocessing

- Bandpass Filtering
- ICA + IC Label
- z-score normalization

3



Windowing
Short windows
(e.g., 2–10 s)
with overlap

4



Encoding

- OPM
- CTM

5



Motif Discovery
Frequent patterns
(motifs) above
support threshold

6

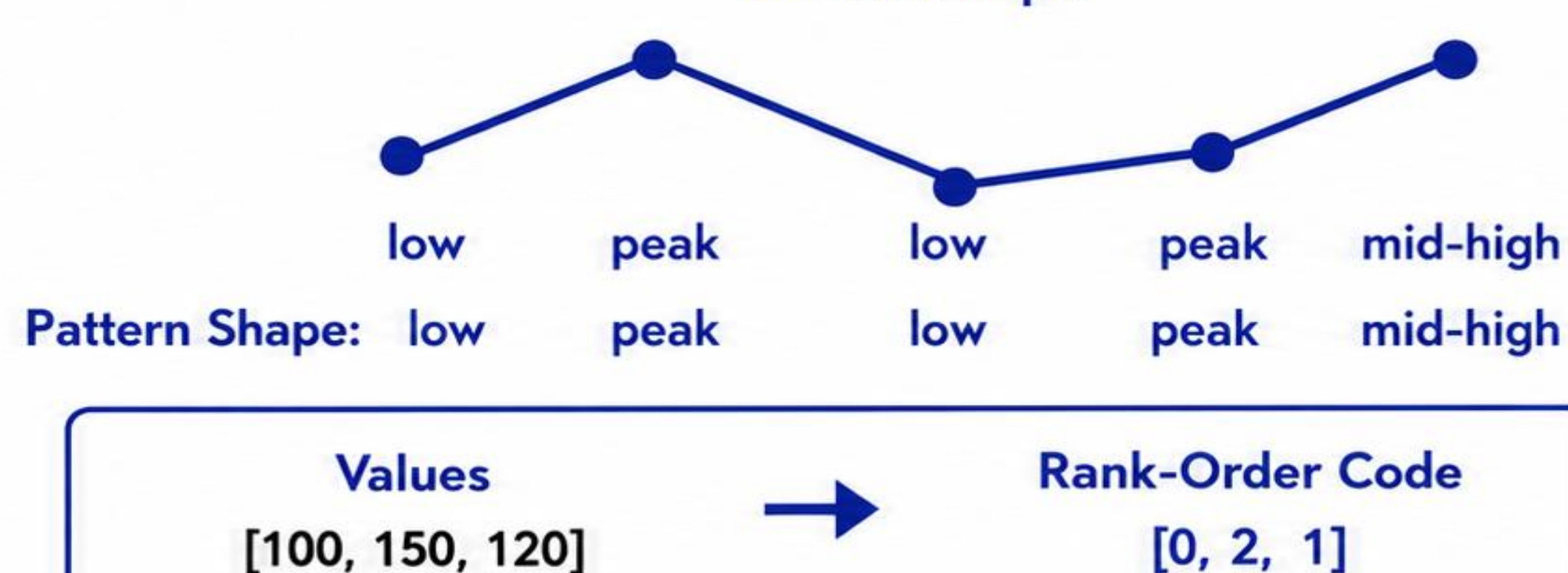


Analysis
Compare motifs &
diversity across
age groups

5. ENCODING METHODS

OPM — Order Preserving Matching

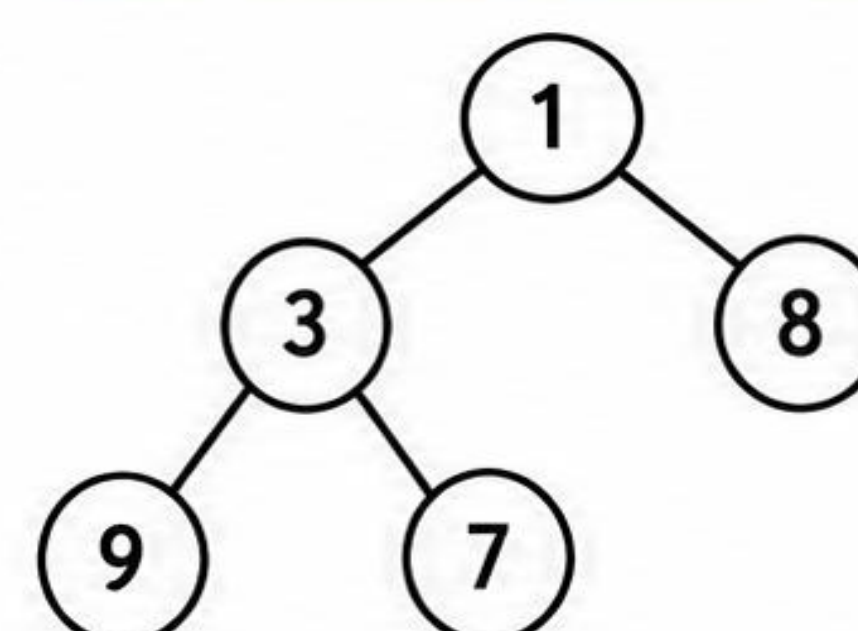
OPM example



CTM — Cartesian Tree Matching

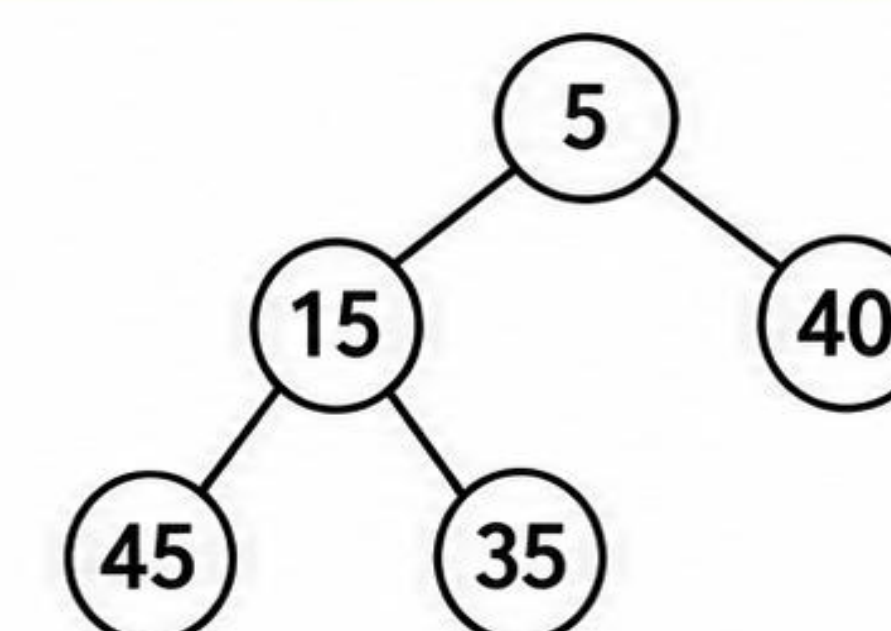
Signal A [9, 3, 7, 1, 8]

Signal B [45, 15, 35, 5, 40]



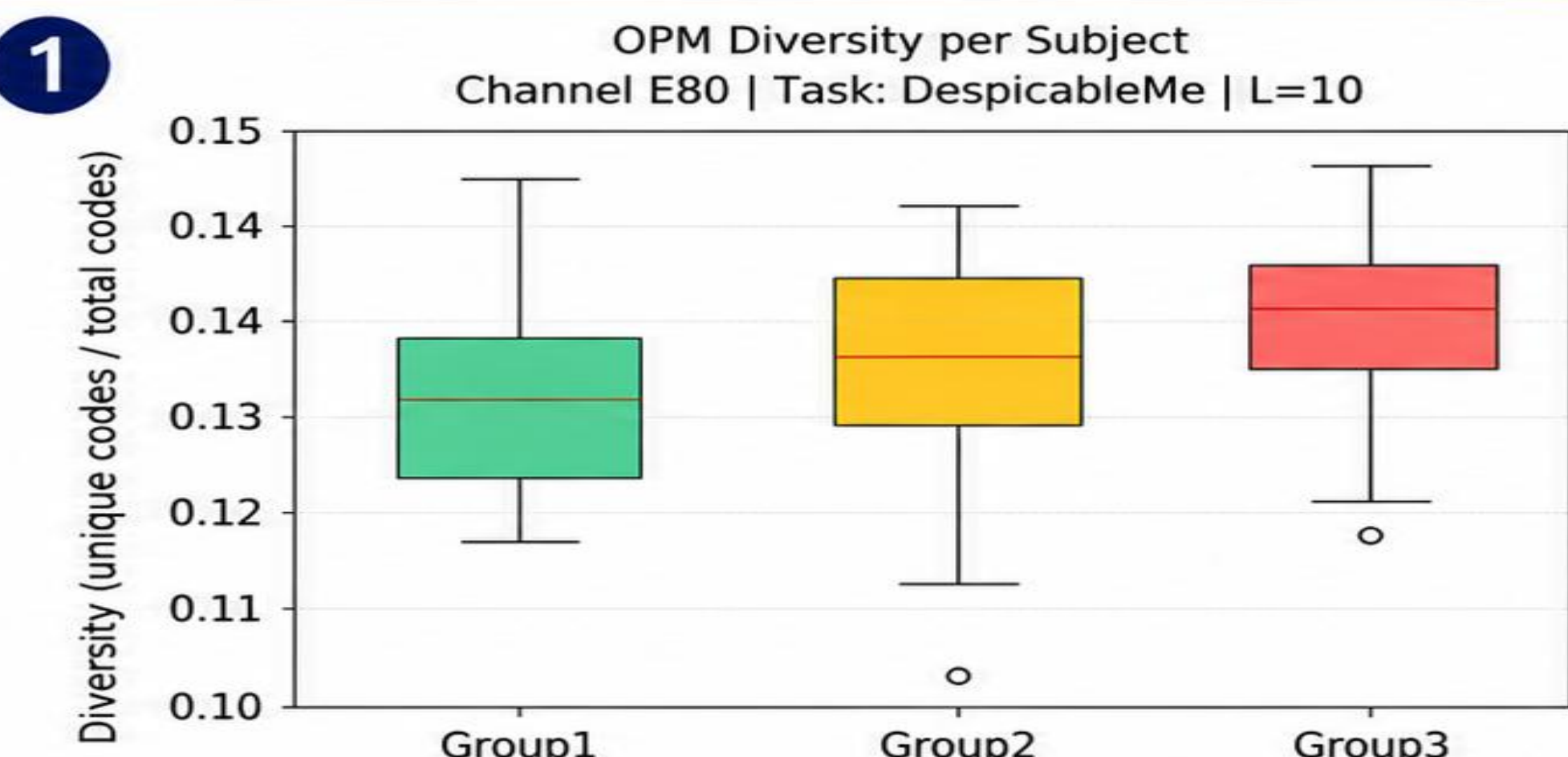
Identical Cartesian Tree Structure

=

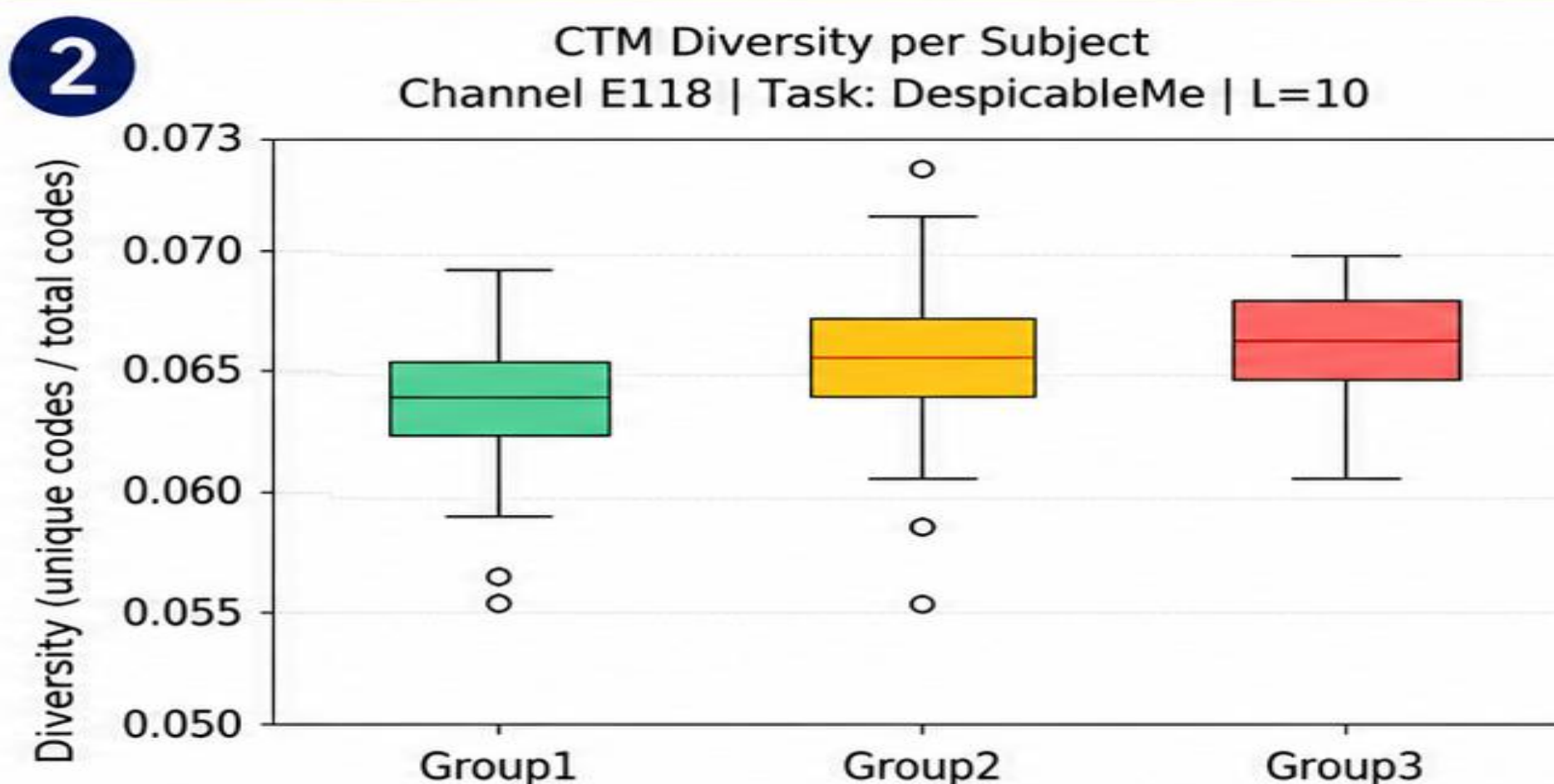


Structural Match

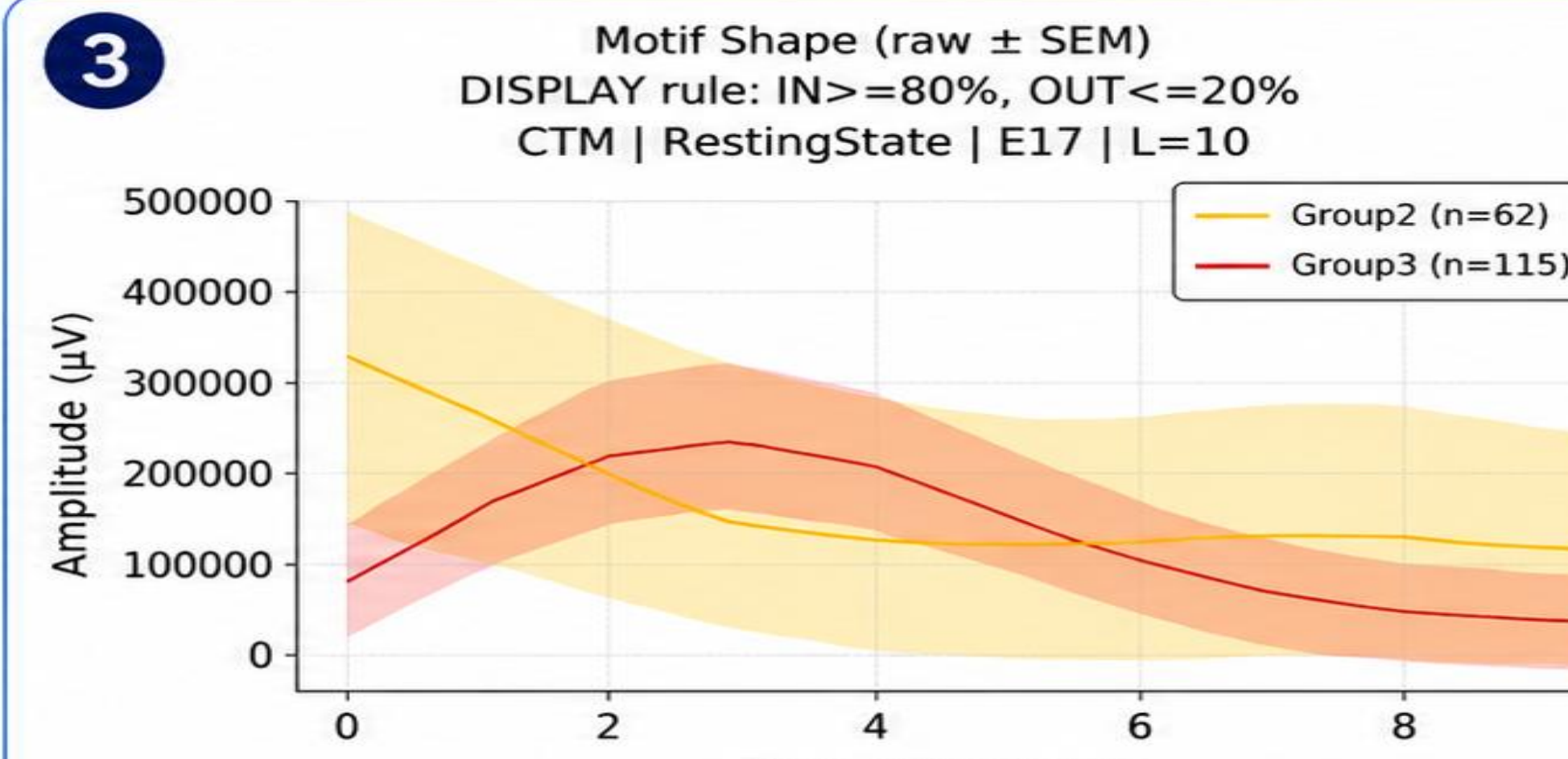
6. RESULTS: REPRESENTATIVE FINDINGS



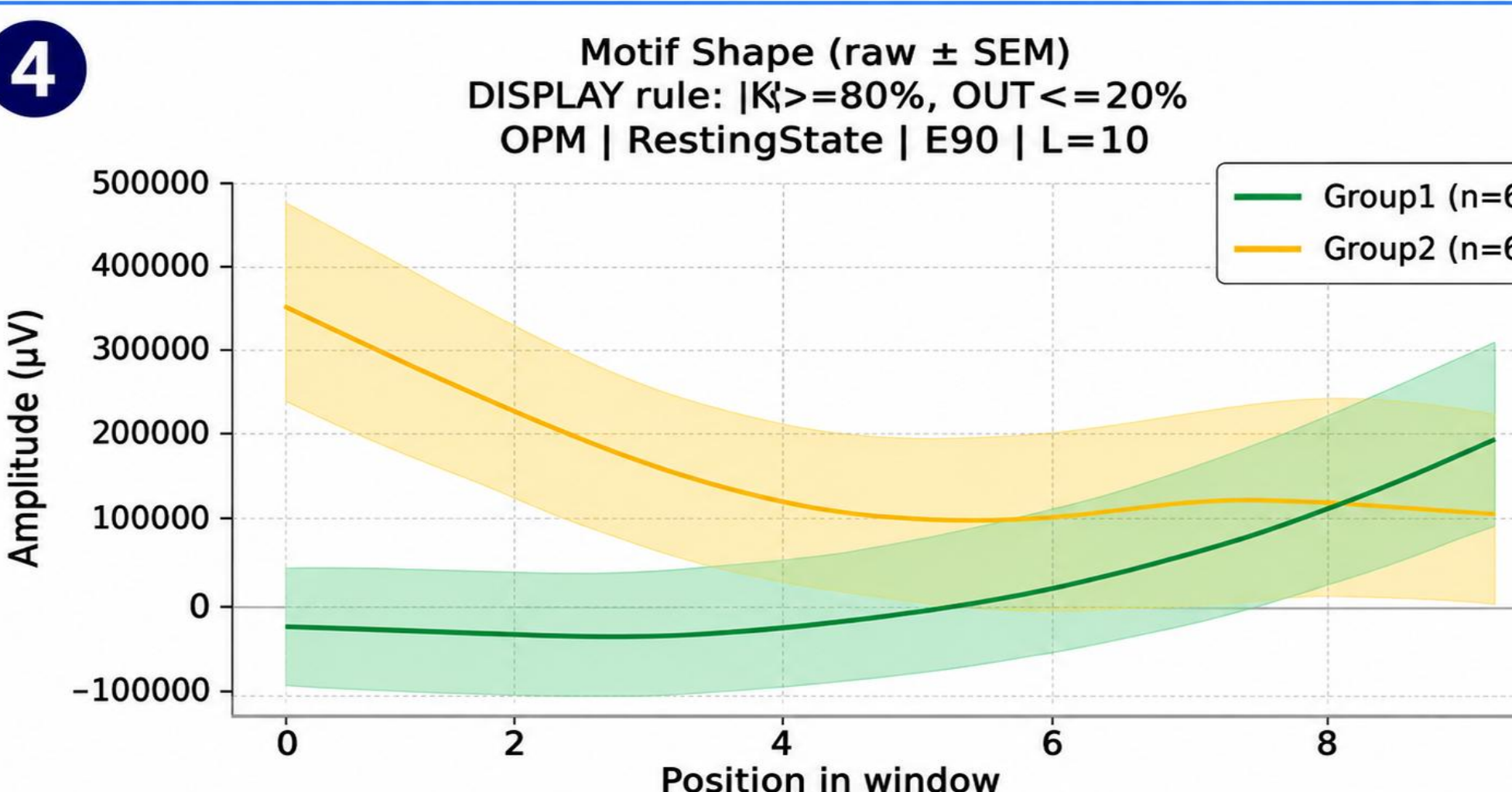
Group3 shows the highest median OPM diversity, while Group1 is lowest. This suggests highly nuanced ordered EEG variation in older participants in this task.



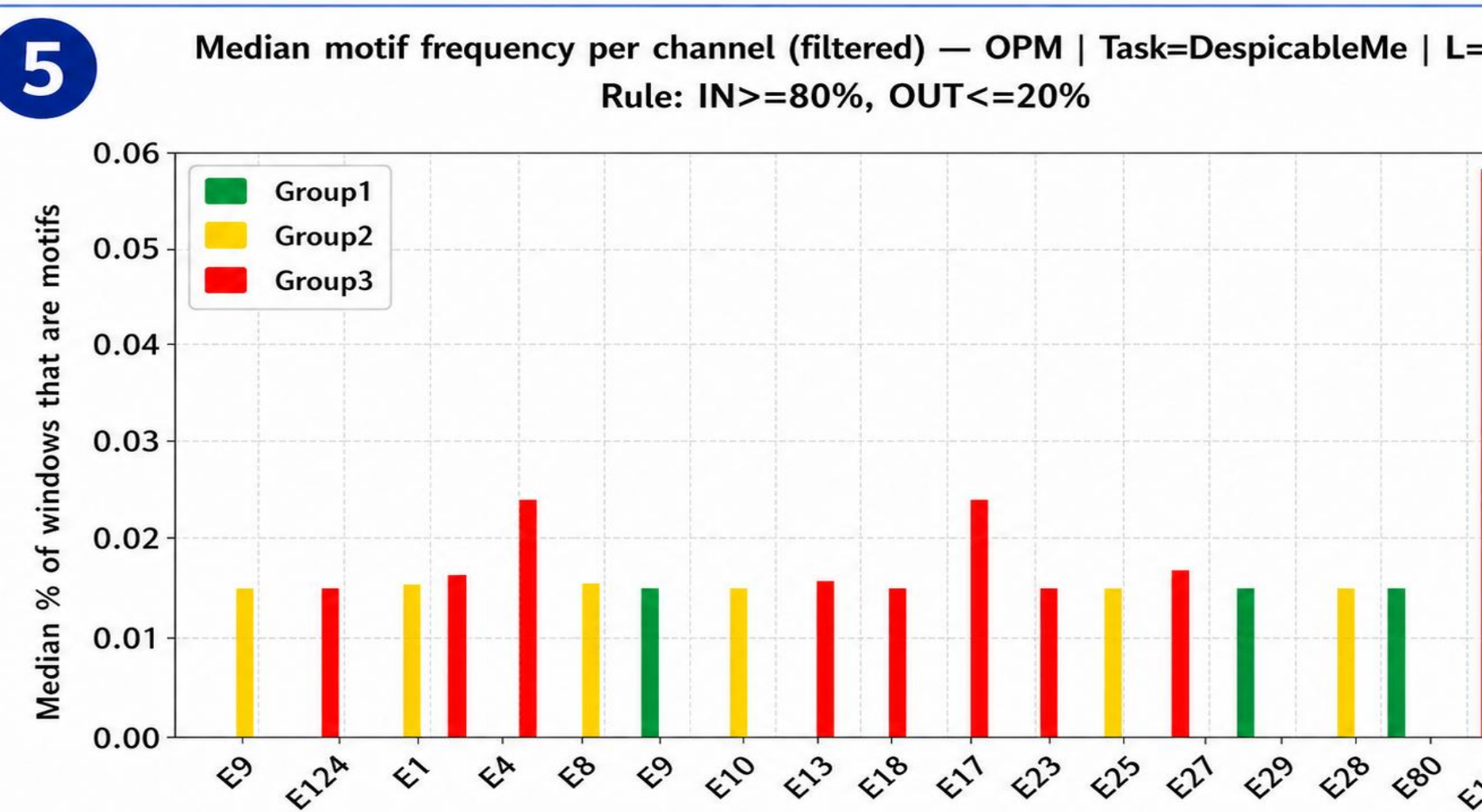
CTM diversity also rises from Group1 to Group2 to Group3. Older participants therefore show more varied global waveform structures in this channel.



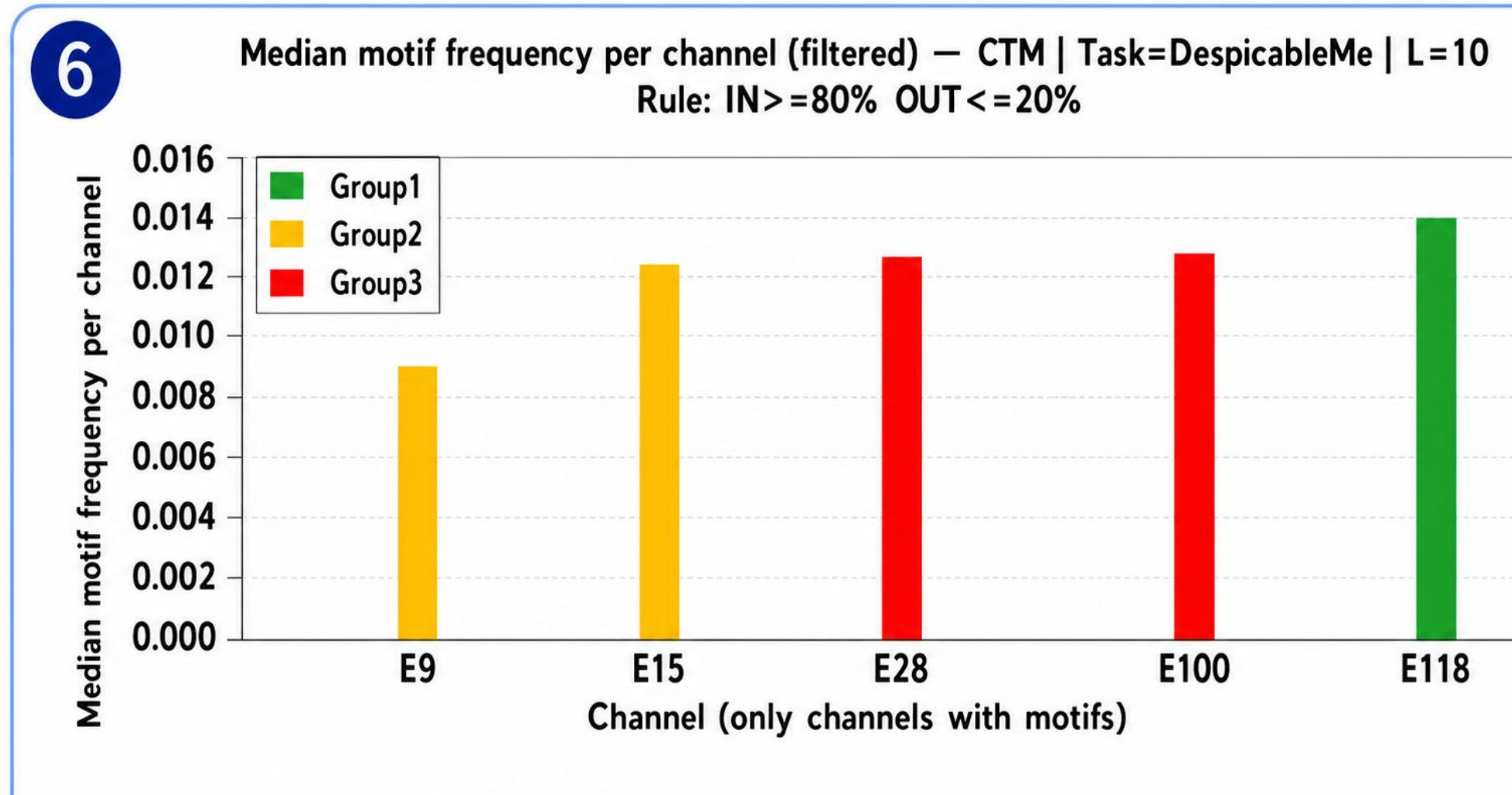
The same CTM motif appears with clearly different average shapes. This indicates task- and channel-related differences in the structure of recurring motifs.



The OPM motif differs clearly between Group1 and 2 across the window. Group1 rises strongly at the end, whereas Group2 starts high and decreases.



Several channels are dominated by motif from a single age group. This shows the local-order motifs are not distributed uniformly across development.



Only a few channels, such as CTM motifs at E100 and E118, tend to be age-specific. This suggests CTM captures more selective age-related structural differences.

7. CONCLUSIONS



Diversity is higher in older participants in some channels, suggesting more varied non-EEG patterns with age.



Motif-shape plots show that the same motifs can have different average shapes across age groups, indicating developmental differences in brain activity footprints.



Motif-frequency results show that some channels are dominated by different age groups, so motif occurrence is not uniform across development.



OPM and CTM provide complementary views, so together they support the existence of shape-based EEG pattern matching for studying normal brain development.



GitHub: github.com/wadlea/EEG-age-patterns

Keywords:

EEG • Brain Development • Motif Discovery • OPM • CTM • HBN-EEG • BIDS